Gianluigi Lopardo

Data Scientist, PhD

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Professional Experience

Jul 2025 – ongoing **Research Analyst** International Policy Analysis Division European Central Bank (Frankfurt, Germany)	*Research and development of machine learning methods for macroeconomic forecasting and policy analysis, exploiting non-traditional data sources* • Leading oil price forecasting project using LLMs to analyze market reports and developing econometric framework for forecast evaluation • Developing AI tools to strengthen the division's analytical framework • Providing analytical support to multiple division projects, *e.g.*, the analysis of US tariff threats and cross-border payment costs with stablecoins
Oct 2024 – May 2025 **PhD Trainee** International Policy Analysis Division European Central Bank (Frankfurt, Germany)	*Research and development of machine learning methods for financial markets and policy analysis, exploiting non-traditional data sources* • Led research project developing firm-level AI exposure measures using textual analysis, examining impact on financial performance • Co-authored ECB Working Paper (under journal submission) and VoxEU column, presenting findings at internal seminars • Proposed and developed *Trump Tariff Threats Index* from analysis of tweets, subsequently integrated into division's policy analysis toolkit
Sep 2021 – Aug 2024 **Teaching Assistant** Université Côte d'Azur (Nice, France)	*Led theoretical and practical instruction in mathematics and statistics* • Delivered lectures for undergraduate and graduate courses in French, including students from economics disciplines • Supervised labs, developed teaching materials, managed assessments
Mar 2021 – Sep 2021 **AI Research Intern** Maasai Team, Inria (Sophia Antipolis, France)	*Research on explainable AI for business decision-making* • Developed the *SMACE* interpretability method and Python package for composite decision systems, resulting in a first-authored publication
Sep 2020 – Feb 2021 **ML Engineer Intern** Alten (Sophia A., France)	*R&D of computer vision solutions to automate document processing* • Developed and optimized computer vision pipelines to automate document digitization and information extraction

Education

Sep 2021 – Sep 2024 **Ph.D. in Applied Mathematics** Inria & Université Côte d'Azur (Nice, France)	• Thesis: *Foundation of Machine Learning Interpretability* • Supervisors: Prof. Damien Garreau and Prof. Frédéric Precioso • In-depth mathematical research on interpretability methods for machine learning models and algorithms • Developed novel interpretability approaches resulting in high-quality publications in top venues, such as ICML, AISTATS, and ECML
Oct 2019 – Oct 2021 **M.Sc. Mathematical Engineering** Politecnico di Torino (Turin, Italy)	• Thesis: *Explainable AI for business decision-making* • Supervisors: Prof. Elena Maria Baralis (PoliTo), Prof. Damien Garreau and Prof. Frédéric Precioso (UCA/Inria), Dr. Greger Ottosson (IBM) • M.Sc. focus on statistical data science, mathematical modeling and optimization, machine learning, risk management, financial engineering
Oct 2016 – Oct 2019 **B.Sc. in Applied Mathematics** Politecnico di Torino (Turin, Italy)	• Thesis: *Quantitative and qualitative analysis on the relation between innovation and economic performances in companies* • Supervisor: Prof. Federico Caviggioli (PoliTo) • B.Sc. focus on core applied mathematical areas including probability, statistics, scientific computing, and programming

Publications

Peer-Reviewed

Lopardo, G., Precioso, F., Garreau, D., *Attention Meets Post-hoc Interpretability: A Mathematical Perspective*, Proceedings of the 41st International Conference on Machine Learning (ICML), 2024

Lopardo, G., Precioso, F., Garreau, D., *A Sea of Words: An In-Depth Analysis of Anchors for Text Data*, Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS), 2023

Lopardo, G., Garreau, D., Precioso, F., Ottosson, G., *SMACE: A New Method for the Interpretability of Composite Decision Systems*, European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2022

Lopardo, G., Precioso, F., Garreau, D., *Understanding Post-hoc Explainers: The Case of Anchors*, 54es Journées de Statistique (JdS), Société Française de Statistique, 2023

Lopardo, G. and Garreau, D., *Comparing Feature Importance and Rule Extraction for Interpretability on Text Data*, 2nd Workshop on Explainable and Ethical AI (XAIE @ ICPR), 2022

Working Papers & Policy Publications

Ca' Zorzi, M., Lopardo, G., Manu, A.-S., *Verba Volant, Transcripta Manent: What Corporate Earnings Calls Reveal About the AI Stock Rally*, European Central Bank Working Paper Series No. 3093, 2025

Ca' Zorzi, M., Lopardo, G., Manu, A.-S., *What Corporate Earnings Calls Reveal About the AI Stock Rally*, VoxEU column, Centre for Economic Policy Research, 2025

Lopardo, G., Precioso, F., Garreau, D., *Faithful and Robust Local Interpretability for Textual Predictions*, arXiv preprint, 2023

Theses

Lopardo, G. *Foundations of Machine Learning Interpretability*, PhD thesis, Université Côte d'Azur, 2024

Lopardo, G. *Explainable AI for business decision-making*, MSc thesis, Politecnico di Torino, 2021

Professional Activities

Awards	Young Researcher Prize 2025 by Nice CdA for "high quality" PhD thesis; 3^{rd} team at ESCB hackathon *From News to Forecast: Experimenting with AI Time Series Models*
Selected Presentations	Presented research at international ML conferences including ICML 2024 (Vienna), AISTATS 2023 (Valencia), ECML 2022 (Grenoble), XAIE @ ICPR 2022 (Montreal); invited talks at Cognizant AI (remote), Maasai (Sophia Antipolis), AI4Media (Florence), and internal ECB seminars (IPA Economic Seminar, AI in Economics Workshop, ML Community Meeting); PhD (Oct 2024) and MSc (Oct 2021) defense
Academic	Reviewer for AISTATS, ECML, KGML workshop; co-organized NWI Workshop
Open Source Projects	`forecast-evaluation` (Nov 2025): Python library for comparing forecast accuracy using statistical tests; Corporate Talks (Aug 2025): Dashboards tracking discussion trends in S&P 500 earnings calls, with GenAI discussions classified by sentiment; Hack the Act! (Feb 2025): RAG chatbot for navigating the European Union AI Act
Additional Experience	Visionary (2019-2022, Turin, Italy): Data-driven analysis and optimization for non-profit event operations; JEToP (2016-2018, Turin, Italy): Led IT projects and development for student-run Junior Enterprise; Deputy Head of IT from Nov 2017

Technical skills

Languages	Italian (native), English (fluent), French (proficient)
Programming	Python (advanced), R, MATLAB (proficient), Julia, Stata, SQL, C, Java (intermediate)
ML & NLP	LLMs (OpenAI, Azure, Ollama), time-series foundation models (Chronos, Moirai), retrieval-augmented generation (LangChain, vector DBs), transformer architectures (PyTorch, HuggingFace), interpretability, sentiment analysis, topic modeling
Econometrics & Statistics	Time-series analysis and forecasting, evaluation, risk quantification, causal inference (DiD, panel methods, PSM), statistical inference and pre-asymptotic statistics